On Variance Reduction in Learning Mean Flows

Juanwu Lu
Purdue University
juanwu@purdue.edu

Ziran Wang
Purdue University
ziran@purdue.edu

Abstract

Mean Flows promise distillation-free, one-step generation by optimizing an identity between the average and instantaneous velocity field. In practice they suffer from a non-decreasing, high-variance loss whose root cause has been diagnosed by concurrent work as gradient conflict, Jacobian amplification, and a curvature bottleneck. We show that all three diagnoses are manifestations of a single quantity—the material derivative of the marginal velocity field—and call this the *curvature-variance principle*. From this principle we derive *Variance-Aware MeanFlow* (VaMF), a backbone-preserving training framework that combines an EMA velocity anchor (deterministic JVP tangent), a flow-matching anchor (boundary supervision), and a Hutchinson-estimated trace weight that downweights each sample by the local Jacobian-deviation norm $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$—a closed-form scalar approximation to the BLUE per-sample weight under isotropic conditional-velocity noise.

Add experiment results later...

1 Introduction

Deep generative models that leverage deep neural networks to learn and generate samples from unknown data distributions have achieved profound success. One of the predominant paradigms adopts flow-based generative models [1, 2, 3, 4, 5, 6, 7] that learn continuous-time transformations between a simple prior and a target data distribution. The existing conditional flow matching [5, 6] and stochastic interpolant [7, 8] enable simulation-free training by regressing a velocity field, achieving sample quality competitive with diffusion models [9, 10, 11, 12, 13, 14, 15] and variational autoencoders [16]. However, flow-based models share the same limitation as diffusion models as they require iterative integration at inference time, which can suffer from extensive computational costs and numerical instability.

The issue motivates consecutive studies on reducing the number of function evaluations to one or few steps [17]. Existing approaches include progressive distillation [18], distribution matching [19, 20], consistency models [21, 22], and flow map matching [23], but most require a pre-trained teacher model. Recently, Mean Flows [24] offers an appealing *distillation-free* alternative by learning a two-parameter average velocity field $\boldsymbol{u}(\boldsymbol{x}, r, t)$ through a total-derivative identity between the average and marginal velocity field. In principle, this identity provides exact self-supervision. The compound prediction $\boldsymbol{u} + (t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}$ should match the instantaneous velocity at convergence, requiring no external teacher. Nevertheless, empirical training of Mean Flows is plagued by non-decreasing loss and prohibitively high gradient variance [25, 26].

In this work, we conduct a theoretical analysis to showcase that these issues are *three manifestations of a single quantity*: the material derivative $\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} := \partial_t \boldsymbol{v} + (\nabla_{\mathbf{x}}\boldsymbol{v})\boldsymbol{v}$ of the marginal velocity field, which measures how rapidly the velocity changes along its own trajectories. We call this the **curvature-variance principle**. Specifically, we prove that: **(i)** the curvature gap $\Delta := \boldsymbol{u} - \boldsymbol{v}$ satisfies $\Delta \approx -\frac{(t-r)}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$, so it is the local flow acceleration that causes the average and instantaneous velocities to diverge; **(ii)** the per-step gradient variance is amplified by a Jacobi factor $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ where $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$, and the stop-gradient operator creates a semi-gradient gap that blinds the optimizer to this amplification; and **(iii)** the acceleration-to-noise ratio $\mathrm{ANR} = (t-r)\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|/\sqrt{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})}$ governs the optimal sampling distribution over intervals, connecting scheduling heuristics in prior work to a principled criterion.

From this analysis we derive **Variance-Aware MeanFlow (VaMF)**, a training framework that addresses all identified failure modes without architectural modification. First, we use an *EMA velocity anchor* in the total derivative to establish a deterministic JVP tangent, provably eliminating Jacobian variance amplification. We then introduce a *flow matching anchor*, an additional objective that supervises the boundary condition and prevents tangent bias. Finally, we apply a *trace weight* that downweights each sample by the local Jacobian-deviation norm $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$, which is a closed-form scalar approximation, estimated with a single Hutchinson probe, to the Best Linear Unbiased Estimator (BLUE) per-sample weight under the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$. The trace weight subsumes inverse-variance weighting, progressive scheduling, and acceleration-aware interval sampling in a single per-sample scalar, while preserving the original MSE objective in Mean Flows. In summary, our contributions in this paper are as follows:

C1 We establish a theoretical principle that unifies gradient conflict, Jacobian amplification, and the curvature bottleneck under the curvature-variance principle, including the first formal quantification of the semi-gradient gap in learning Mean Flows.

C2 We derive the VaMF training framework that eliminates variance amplification and tangent bias through introducing EMA anchoring, FM anchoring, and a Hutchinson-estimated trace weight, without modifying the backbone.

C3 Our experiments showcase that VaMF can significantly improve the both the convergence efficiency and the performance of the original Mean Flows.

A brief summary of the quantity

.

2 Preliminary

Flow-Matching. Let $\mathcal{X} \subset \mathbb{R}^d$ be the space of random variables $\mathbf{x} \in \mathbb{R}^d$ and $\boldsymbol{x}$ a data sample as the realized state. The objective of flow-based generative models is to learn a time-dependent diffeomorphic map $\psi(\boldsymbol{x}, t) : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ from a simple distribution $\mathbf{x}_1 \sim p(\boldsymbol{x}, 1)$ to the target data distribution $\mathbf{x}_0 \sim p(\boldsymbol{x}, 0) = p_{\mathrm{data}}$. Specifically, this is achieved by learning a *time-dependent velocity field* $\boldsymbol{v}(\boldsymbol{x}, t) : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$:

$$\boldsymbol{v}(\boldsymbol{x}, t) \triangleq \frac{\mathrm{d}}{\mathrm{d}t}\psi(\boldsymbol{x}, t), \quad \psi(\boldsymbol{x}, 0) = \boldsymbol{x}_0. \tag{1}$$

The map $\psi(\boldsymbol{x}, t)$ induces a *time-dependent probability density path* $p(\boldsymbol{x}, t) : \mathbb{R}^d \times [0,1] \to \mathbb{R}_{>0}$, with $\int_{\mathcal{X}} p(\boldsymbol{x}, t)\mathrm{d}\boldsymbol{x}$, $\forall t \in [0,1]$. A sufficient and necessary condition for a vector field to generate a probability path is the *continuity equation*, which is a partial differential equation (PDE) that writes

$$\frac{\partial}{\partial t} p(\boldsymbol{x}, t) + \mathrm{div}\left(p(\boldsymbol{x}, t)\boldsymbol{v}(\boldsymbol{x}, t)\right) = 0, \tag{2}$$

where $\mathrm{div}(f)$ denotes the divergence of function f. However, the exact *marginal velocity field* $\boldsymbol{v}(\boldsymbol{x}, t)$ is generally *inaccessible*, especially in a high-dimensional sample space. The optimal-transport conditional flow-matching (OT-CFM) [5, 6] alternatively learns a conditional velocity field $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0; \boldsymbol{\theta}) \triangleq \boldsymbol{v}\left((1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1; \boldsymbol{\theta}\right)$ There exists a relationship between the marginal and conditional velocity field given by the following lemma:

Lemma 1. *Given vector fields $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$ that generate probability paths $p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$, for any data distribution $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, the marginal velocity field is equal to the expected conditional velocity field*

$$\boldsymbol{v}(\boldsymbol{x}, t) = \mathbb{E}_{\boldsymbol{x}_0 \sim p(\boldsymbol{x}_0, t \mid \boldsymbol{x})}\left[\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\right]. \tag{3}$$

See Appendix B for its proof.

One-step Mean Flows. The issue with diffusion [11, 13] and flow-matching models [5] is the integration. Evaluating the velocity field multiple times is not only expensive but can also lead to numerical instability. A Mean Flow model [24] instead learns a two-parameter average velocity field $\boldsymbol{u}(\mathbf{x}, r, t)$ to enable efficient sampling. It leverages the identity between the average and marginal velocity field given by

$$(t-r)\boldsymbol{u}(\boldsymbol{x}, r, t) = \int_{\tau=r}^{t} \boldsymbol{v}(\boldsymbol{x}, \tau)\mathrm{d}\tau, \tag{4}$$

where r and t are scalars, the start and end time steps along the probability density path. Taking the total derivative with respect to the terminal time step t yields

$$\boldsymbol{u}(\boldsymbol{x}, r, t) + (t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}(\boldsymbol{x}, r, t) = \boldsymbol{v}(\boldsymbol{x}, t), \tag{5}$$

where the total derivative of the average velocity field with respect to t can be expressed as a Jacobi-vector Product (JVP)

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}(\boldsymbol{x}, r, t) = \partial_{\boldsymbol{x}}\boldsymbol{u} \cdot \boldsymbol{v}(\boldsymbol{x}, t) + \partial_r \boldsymbol{u} \cdot \frac{\mathrm{d}r}{\mathrm{d}t} + \partial_t \boldsymbol{u} \cdot \frac{\mathrm{d}t}{\mathrm{d}t} = \partial_{\boldsymbol{x}}\boldsymbol{u} \cdot \boldsymbol{v}(\boldsymbol{x}, t) + \partial_t \boldsymbol{u} \tag{6}$$

We abuse the notation and express the right-hand side by $\mathtt{JVP}(\boldsymbol{u}, (\boldsymbol{x}, r, t), (\boldsymbol{v}(\boldsymbol{x}, t), 0, 1))$. In the original MeanFlow [24], the authors replace both the marginal velocity field in the right-hand side of equation 5 and inside the JVP with the conditional velocity field, and derive the MeanFlow loss function given by

$$\mathcal{L}_{\mathrm{MF}}(\boldsymbol{\theta}) = \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\mathtt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{z}, r, t), (\boldsymbol{v}_{\mathrm{cond}}, 0, 1))\right] - \boldsymbol{v}_{\mathrm{cond}}\|_2^2\right], \tag{7}$$

where $r \sim \mathcal{U}(0,1)$, $t \sim \mathcal{U}(0,1)$, $r \le t$, $\boldsymbol{x}_0 \sim p_{\mathrm{data}}$, $\boldsymbol{x}_1 \sim \mathcal{N}(0, \boldsymbol{I}_d)$, $\boldsymbol{z} = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$, $\boldsymbol{v}_{\mathrm{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$, and $\mathtt{sg}[\cdot]$ is the stop-gradient operator. The authors highlight that stopping gradients in JVP helps eliminate the need for "double backpropagation" and high-order derivatives.

3 Variance-aware Mean Flows

The MeanFlow loss in equation 7 is empirically known to be *non-decreasing* and to suffer from *high-variance gradients* [25, 26]. In this section, we aim to answer the two complementary questions:

***RQ1* (Diagnosis): What ist the root cause for the training pathologies in Mean Flows?**

Existing works diagnose MeanFlow's instability from three apparently distinct angles, such as gradient conflict [25] and the curvature bottleneck [27]. Are they symptoms of a single underlying mechanism, and if so, what observable in the gradients does that mechanism affect?

***RQ2* (Prescription): Variance-aware loss design.**

Given the identified mechanism, what is the simplest modification that can drive the gradient variance toward its irreducible level?

In the following, section 3.1 answers *RQ1* at the level of the gradients. We apply Reynolds decomposition to reveal both the bias induced by stop-gradient training and the per-step gradient variance governed by the Jacobi factor $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$, and establish a connection between the bias and the *curvature gap* $\Delta = \boldsymbol{u} - \boldsymbol{v}$. Section 3.2 then quantifies the gap in terms of flow acceleration with a Taylor expansion, exposing the material derivative as its physical source of the curvature gap. Section 3.3 answers *RQ2* by proposing our training framework, *Variance-Aware MeanFlow (VaMF)*, comprising an EMA velocity anchor, a flow-matching anchor, and a closed-form trace weight, all applied on top of the original MSE objective without any architectural change. Full proofs are deferred to Appendix C.

3.1 Bias and Variances in Learning Mean Flows

We begin with investigating the consequences of replacing the inaccessible marginal velocity $\boldsymbol{v}(\boldsymbol{x}, t)$ by the conditional field $\boldsymbol{v}_{\mathrm{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$ in both the JVP tangent and the regression target. We first apply the Reynolds decomposition of the field and let $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0) = \boldsymbol{v}(\boldsymbol{x}, t) + \boldsymbol{v}'$, where $\boldsymbol{v}'$ is a zero-mean fluctuation from the marginal velocity induced by optimal transport between latent $\boldsymbol{x}_1$ and data $\boldsymbol{x}_0$. The regression loss equation 7 then expands to

$$\begin{aligned}\mathcal{L}_{\mathrm{MF}}(\boldsymbol{\theta}) &= \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}_{\mathrm{cond}} + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}\right] - \boldsymbol{v}_{\mathrm{cond}}\|_2^2\right] \\ &= \mathbb{E}_{r,t,\boldsymbol{x}_t,\boldsymbol{x}_1}\left[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\left[\left\|\underbrace{\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}(\boldsymbol{x}_t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}\right] - \boldsymbol{v}(\boldsymbol{x}_t, t)}_{\text{mean-field residual } \boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}(\boldsymbol{x}_t, t)} + \mathtt{sg}[\boldsymbol{J}]\boldsymbol{v}'\right\|_2^2\right]\right].\end{aligned} \tag{8}$$

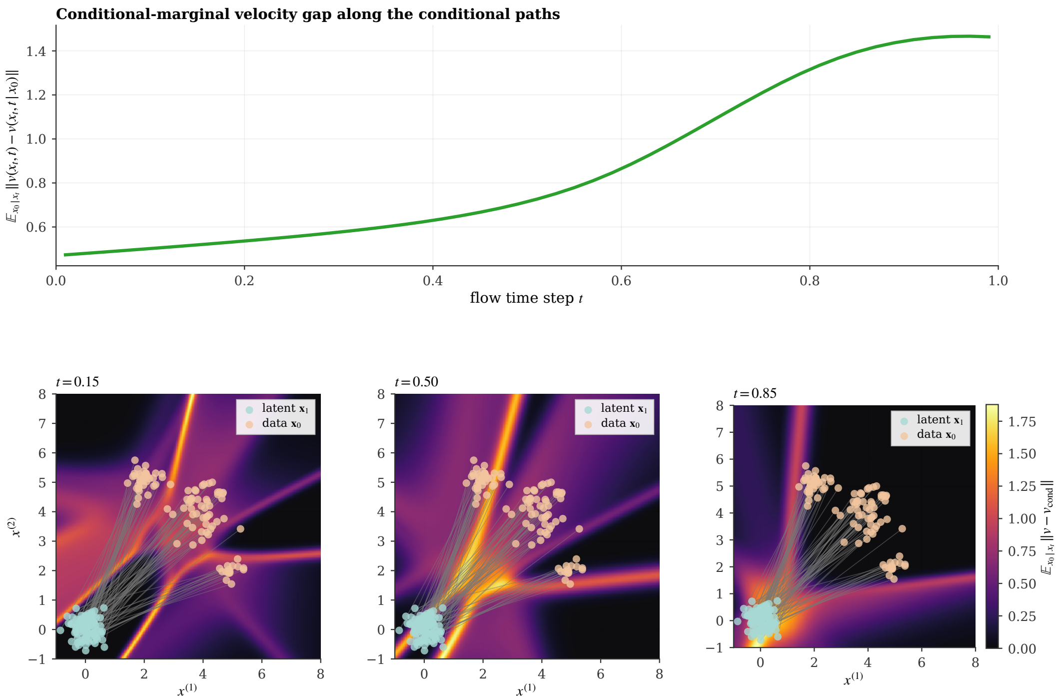

Figure 1: Conditional-marginal velocity gap on a 3-mode mixture. **Top:** the expected per-sample fluctuation $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\|\boldsymbol{v}(\boldsymbol{x}_t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t \mid \boldsymbol{x}_0)\|$, evaluated along the conditional paths, grows monotonically with t as conditional trajectories disperse, with the steepest rise where $\boldsymbol{x}_t$ traverses the mode-mixing region. **Bottom:** the spatial heatmap at three timesteps confirms that the gap is heterogeneous and concentrates in the mode-mixing regions where multiple conditional paths overlap.

Herein, we abuse the notations and let $\boldsymbol{J} \triangleq (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_\theta - \boldsymbol{I}_d \in \mathbb{R}^{d\times d}$ denote a Jacobi factor of $\boldsymbol{u_\theta}$ given sample $\boldsymbol{x}_t = \boldsymbol{x}_1 - \boldsymbol{x}_0$. This decomposition reveals that the original loss for learning Mean Flows is the sum of a deterministic mean-field residual $\boldsymbol{r}^{\text{sg}}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t)$ and a sample-based Jacobi-velocity-product $\text{sg}[\boldsymbol{J}]\boldsymbol{v}'$, where the former is the actual loss we aim to minimize given the identity in equation 5. If we further expand the inner expectation, we have $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = 0$ based on equation 3 and the cross term vanishes, resulting in

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell_{\text{MF}}(\boldsymbol{\theta})] = \mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\left[\|\boldsymbol{r}(\boldsymbol{x}_t, t) + \text{sg}[\boldsymbol{J}\boldsymbol{v}']\|_2^2\right] = \|\boldsymbol{r}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t)\|_2^2 + \text{sg}[\text{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)], \tag{9}$$

where $\Sigma_{\boldsymbol{v}'} = \text{Cov}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}']$ is the *non-zero* covariance of the conditional velocity field and $\ell_{\text{MF}}(\boldsymbol{\theta})$ is the per-sample loss. Therefore, replacing marginal velocity fields wih conditional ones gives birth to an *irreducible per-sample variance* of the conditional fluctuation given by the trace $\text{Tr}(\Sigma_{\boldsymbol{v}'})$, where the matrix $\boldsymbol{J}$ converts it into a per-parameter loss contribution that grows quadratically in the spatial Jacobian. This establishes the following theorem.

Theorem 2 (Jacobian Variance Amplification.)**.** *The Jacobi factor amplifies the variances of the original MeanFlow loss*

$$\boxed{\text{Var}\left[\nabla_{\boldsymbol{\theta}}\ell_{MF}(\boldsymbol{\theta}) \mid \boldsymbol{x}_t\right] \propto \text{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)}$$

Theorem 2 explains the instability and high variance observed in training the MeanFlow model. With large intervals $(t-r)$ or large norm of the Jacobi term $\|\partial_{\boldsymbol{x}_t}u_\theta\|_2^2$, the per-sample gradient is dominated by the noise term $\boldsymbol{J}\boldsymbol{v}'$, downweighting the mean-field signal from $\boldsymbol{r}^{\text{sg}}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t)$.

To help our readers better understand the existence of such a gap, we visualize the magnitude of this fluctuation $\|\boldsymbol{v}'\|$ on a example 3-Gaussian mixture in fig. 1. It is clear that such a gap is *non-zero*, *increases* when approaching the latent, and concentrates spatially in a *mode-mixing regions near the* *latent* where multiple conditional paths overlap. Meanwhile, since the inner expectation includes the Jacobian amplified trace, we find that the stop-gradient operator creates a *semi-gradient gap*.

Theorem 3 (Semi-Gradient Gap)**.** *The gradients of MeanFlow loss with and without the stop-gradient operator differ by*

$$\boxed{\underbrace{2(t-r)\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\nabla_{\boldsymbol{\theta}}\left(\partial_{\boldsymbol{x}_t}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v}(\boldsymbol{x}_t,t)+\partial_t\boldsymbol{u}_{\boldsymbol{\theta}}\right)\cdot\boldsymbol{r}_\theta\right]}_{\textit{mean-field gradient difference}}+\underbrace{\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\nabla_{\boldsymbol{\theta}}\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)\right]}_{\textit{variance-driven gradient difference}}}$$

Herein, the stop-gradient eliminates the trace term from the expected gradient and $\boldsymbol{r}_{\boldsymbol{\theta}} \to \boldsymbol{0}$ at convergence. Therefore, the variance-driven gradient difference dominates the semi-gradient gap near convergence. Without stop-gradient, $\nabla_{\boldsymbol{\theta}}\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ will force the system to minimize $\|(t-r)\partial_{\boldsymbol{x}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d\|$, essentially pushes $\partial_{\boldsymbol{x}}\boldsymbol{u}_{\boldsymbol{\theta}} \to \frac{1}{t-r}\boldsymbol{I}_d$ to minimize the magnitude of $\boldsymbol{J}$. On the contrary, stop-gradient operator blinds variance-driven gradient, and as a result, $\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ can potentially increase faster than the magnitude of decrease in $\|\boldsymbol{r}_{\boldsymbol{\theta}}\|_2^2$, eventually leading to the non-decreasing loss in learning Mean Flows.

To understand the root cause for both failure modes in theorem 2 and theorem 3, we define the *curvature gap* as the difference between the average and instantaneous velocity fields, by rearranging the identity in equation 5:

$$\Delta(\boldsymbol{x}_t,r,t) := \boldsymbol{u}(\boldsymbol{x}_t,r,t) - \boldsymbol{v}(\boldsymbol{x}_t,t) = -(t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u} \tag{10}$$

The mean-field residual in equation 8 can therefore be rewritten as $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}} = \Delta_{\boldsymbol{\theta}} + (t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}_{\boldsymbol{\theta}}$, where we simplify our notation and denote $\Delta_{\boldsymbol{\theta}} := \boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{v}$. At convergence the total derivative should exactly cancel the curvature gap, so the training signal $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is a small difference of two quantities whose magnitudes scale with $\|\Delta\|$. However, when $\|\Delta\|$ is large, this cancellation becomes delicate. Specifically, the gradient signal from $\boldsymbol{r}_{\boldsymbol{\theta}}$ is weak relative to the noise $\boldsymbol{J}\boldsymbol{v}'$, while the stop-gradient operator prevents the optimizer from controlling the resulting variance growth. This raises a natural question: how large is Δ in practice, and what governs its magnitude?

3.2 Flow Acceleration

The curvature gap $\Delta(\boldsymbol{x}_t, r, t)$ defined in equation 10 controls the severity of both Jacobian variance amplification and the semi-gradient gap. We now quantify Δ using flow acceleration, revealing its dependence on the material derivative of the velocity field.

Theorem 4 (Flow Acceleration Expansion of the Expectation Gap)**.** *Let $\boldsymbol{v}(\boldsymbol{x},t) \in C^2$ be twice continouously differentiable. The expectation gap admits a Taylor series at $\boldsymbol{x}_t$:*

$$\Delta(\boldsymbol{x}_t,r,t) = -\frac{(t-r)}{2}\frac{\mathrm{D}v}{\mathrm{D}t}(\boldsymbol{x}_t,t) + \mathcal{R}_2(\boldsymbol{x}_t,r,t), \tag{11}$$

where $\frac{\mathrm{D}v}{\mathrm{D}t}(\boldsymbol{x},t) \triangleq \partial_t\boldsymbol{v}(\boldsymbol{x},t) + (\nabla_{\boldsymbol{x}}\boldsymbol{v}(\boldsymbol{x},t))\cdot\boldsymbol{v}(\boldsymbol{x},t)$ is the material dervative (a.k.a. flow acceleration) and $\mathcal{R}_2(\boldsymbol{x},r,t)$ is the second-order remainder of the Taylor series.

Herein, it is reasonable to assume $\boldsymbol{v}(\boldsymbol{x},t) \in C^2$ since existing work primarily uses a U-Net [28, 12] or transformer backbone with twice continuously differentiable activation functions (*e.g., the GELU activation [29]*) to learn the velocity field. Theorem 4 establishes a theoretical perspective that explains the points discussed in the existing works. Specifically, methods with rectified flows [30] straighten the marginal ODE trajectory, thereby enforcing near-zero flow accelerations. By theorem 4, this pushes $\Delta(\boldsymbol{x},r,t) \to 0$ and eliminates the root cause for training instability. Meanwhile, methods that employ progressive scheduling [26] bypasses the problem by initially training with a small interval $r \to t$. According to theorem 4, the expectation gap grows at a rate factor $\frac{(t-r)}{2}\left\|\frac{\mathrm{D}}{\mathrm{D}t}(\boldsymbol{x},r,t)\right\|$ and hence it is reasonable to constrain the maximum interval during training by bounding this factor.

In addition, the relationship between the expectation gap and the flow acceleration reveals a challenge in the intermediate time step t, where the flow acceleration maximizes due to the intersection of conditional trajectories (as previously shown in fig. 1). We argue that the issue can exacerbated for high-resolution, multi-modal data, where marginal velocity field changes occur rapidly.

3.3 Variance-Aware MeanFlow

We propose a concrete and simple training framework to, Variance-Aware MeanFlow (VaMF), that addresses all identified failure modes above. It address the Jacobian variance amplification through veclocity anchoring (section 3.3.1) and reduces per-sample variance with adaptive weighting (section 3.3.3).

3.3.1 Target Network

In equation 8, the $\boldsymbol{J}\boldsymbol{v}'$ amplification arises because $\boldsymbol{v}'$ enters the JVP tangent, where it multiplies the spatial Jacobian $(t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}$. We notice that, if the tangent were replaced by any *deterministic estimate* $\hat{\boldsymbol{v}}$, the fluctuation $\boldsymbol{v}'$ would only appear in the regression target, entering the residual linearly rather than through $\boldsymbol{J}$. In this case, the per-sample loss reduces to $\|\tilde{\boldsymbol{r}}_{\boldsymbol{\theta}}\|_2^2 + \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$, where $\tilde{\boldsymbol{r}}_{\boldsymbol{\theta}}$ is a deterministic residual. As a result, the variance drops to the irreducible level $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$ without any Jacobian amplification. We higlight this as a theoretical foundation missing from the proposed method in a recent study [26]. The remaining question is whether we can achieve this without learning a separate velocity model.

We propose to replace the stochastic JVP tangent $\boldsymbol{v}_{\text{cond}}$ with a deterministic estimate from a target network. Specifically, we use a Polyak averaging [31] (*a.k.a. exponential moving average (EMA)* [32]) of the model parameters as the target network, and the new loss writes

$$\mathcal{L}_{\text{MF}}^{\text{EMA}}(\boldsymbol{\theta}) = \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\left\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\mathtt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{z}, r, t), (\boldsymbol{u}_{\bar{\boldsymbol{\theta}},t}, 0, 1))\right] - \boldsymbol{v}_{\text{cond}}\right\|_2^2\right], \tag{12}$$

where $\boldsymbol{u}_{\bar{\boldsymbol{\theta}},t} = \boldsymbol{u}(\boldsymbol{x}_t, t, t)$ and $\bar{\boldsymbol{\theta}}$ is updated by $\bar{\boldsymbol{\theta}} \leftarrow \mu\bar{\boldsymbol{\theta}} + (1-\mu)\boldsymbol{\theta}$ with $\mu \in [0, 1)$.

3.3.2 Flow-matching Anchor

The EMA tangent helps resolve the Jacobian variance amplification but introduces a transient bias when $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \neq \boldsymbol{v}(\boldsymbol{x}_t, t)$. To correct this, we add a *flow matching anchor* that provides direct velocity supervision at the boundary $r \to t$:

$$\mathcal{L}_{\text{FM}}(\boldsymbol{\theta}) = \mathbb{E}_{t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\left\|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t-\delta, t) - \boldsymbol{v}_{\text{cond}}\right\|_2^2\right], \tag{13}$$

where $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$ with $\delta_{\max} \ll 1$. For small δ, the average velocity over $[t-\delta, t]$ approximates the instantaneous velocity. The full objective in VaMF is given by

$$\mathcal{L}_{\text{VaMF}}(\boldsymbol{\theta}) = \mathcal{L}_{\text{MF}}^{\text{EMA}}(\boldsymbol{\theta}) + \lambda\,\mathcal{L}_{\text{FM}}(\boldsymbol{\theta}), \tag{14}$$

where $\lambda > 0$ balances the two terms. This establishes the following propositions that characterize the gradient structure.

Proposition 5 (Gradient of VaMF Loss)**.** *The gradient of $\mathcal{L}_{MF}^{EMA}$ takes the form $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{MF}^{EMA} = 2\,\mathbb{E}\left[\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)\cdot\tilde{\boldsymbol{r}}^{EMA}\right]$, where $\tilde{\boldsymbol{r}}^{EMA} := V_{\boldsymbol{\theta}}^{EMA} - \boldsymbol{v}(\boldsymbol{x}_t, t)$ is the mean-field residual under the EMA tangent. We eliminate the conditional velocity variance.*

Proposition 6 (Bias-Variance Tradeoff)**.** *The EMA mean-field residual decomposes as $\tilde{\boldsymbol{r}}^{EMA} = \boldsymbol{r}_{\boldsymbol{\theta}} + (t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t))$, where $\boldsymbol{r}_{\boldsymbol{\theta}}$ is the true MeanFlow residual from equation 8. At convergence, $\boldsymbol{r}_{\boldsymbol{\theta}} \to \boldsymbol{0}$ and the FM anchor drives $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$, giving $\tilde{\boldsymbol{r}}^{EMA} \to \boldsymbol{0}$.*

Our proposition 5 shows that the VaMF gradient has a first-order structure with no Jacobian-mediated variance amplification (cf. theorem 2). Proposition 6 shows the transient bias is bounded by $(t-r)\|\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\|\|\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)\|$ and shrinks as the EMA velocity improves.

3.3.3 Adaptive Weights

Although the EMA tangent removes the Jacobian-amplified variance from the *gradient*, samples with large $\|\boldsymbol{J}\|$ still produce loss values that dwarf samples in flat, low-curvature regions, biasing the empirical-risk estimator toward the noisy regime. We addresses this by *adaptively weighting* each sample to equalize its contribution to the expected loss.

We derive the adaptive weight using the Gauss-Markov Theorem. In OT-CFM, we argue it is fair to treat the conditional fluctuation $\boldsymbol{v}'$ as Gaussian noise, and derive the Best Linear Unbiased Estimator

Algorithm 1 Variance-Aware MeanFlow Training

Require: Dataset $\mathcal{D}$, EMA decay μ, anchor weight λ, anchor interval $[\delta_{\min}, \delta_{\max}]$
Require: Schedule σ_t, number of probes K
Initialize $\bar{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}$
for $k = 1, 2, \dots$ **do**
Sample $(\boldsymbol{x}_0, \boldsymbol{x}_1) \sim \mathcal{D} \times \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$, $t \sim \mathcal{U}[0,1]$, $r \sim \mathcal{U}[0,t]$, $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$
$\boldsymbol{x}_t \leftarrow (1-t)\boldsymbol{x}_0 + t\,\boldsymbol{x}_1$, $\quad \boldsymbol{v}_{\text{cond}} \leftarrow \boldsymbol{x}_1 - \boldsymbol{x}_0$
$\boldsymbol{v}_{\text{tang}} \leftarrow \mathtt{sg}[\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)]$ ▷ EMA tangent (D1)
$V_{\boldsymbol{\theta}} \leftarrow \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t) + (t-r)\,\mathtt{sg}[\mathtt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{x}_t, r, t), (\boldsymbol{v}_{\text{tang}}, 0, 1))]$
Sample K probes $\{\boldsymbol{z}_i\}_{i=1}^{K} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$
$\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \leftarrow \frac{1}{K}\sum_{i=1}^{K} \big\|(t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)\,\boldsymbol{z}_i - \boldsymbol{z}_i\big\|_2^2$
$w \leftarrow 1 \,/\, \big(1 + \sigma_t^2\,\mathtt{sg}[\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}]/d\big)$ ▷ trace weight (D3)
$\ell_{\text{MF}} \leftarrow w \cdot \|V_{\boldsymbol{\theta}} - \boldsymbol{v}_{\text{cond}}\|_2^2$
$\ell_{\text{FM}} \leftarrow \|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t-\delta, t) - \boldsymbol{v}_{\text{cond}}\|_2^2$ ▷ FM anchor (D2)
$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\nabla_{\boldsymbol{\theta}}(\ell_{\text{MF}} + \lambda\,\ell_{\text{FM}})$
$\bar{\boldsymbol{\theta}} \leftarrow \mu\bar{\boldsymbol{\theta}} + (1-\mu)\boldsymbol{\theta}$ ▷ EMA update
end for
return $\boldsymbol{\theta}$ ▷ Generate: $\hat{\boldsymbol{x}}_0 = \boldsymbol{x}_1 - \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_1, 0, 1)$

(BLUE) of the mean-field residual $\tilde{\boldsymbol{r}}^{\text{EMA}}$ with the inverse-covariance matrix weight $(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)^{-1}$. However, it further introduce two practical pathologies. First, the per-sample $d\times d$ matrix inversion has a cubic-cost $\mathcal{O}(d^3)$ and is intractable for high-dimensional data or latents. Meanwhile, the matrix is ill-conditioned whenever $(t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \to \boldsymbol{I}_d$ and $\boldsymbol{J} \approx \mathbf{0}$, which pushes the inverse to $+\infty$.

To address these pathologies, we propose to use isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$. Because $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top) = \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$, we define the per-sample *trace weight*

$$w(\boldsymbol{x}_t, r, t) \;:=\; \frac{d}{1 + \sigma_t^2\,\mathtt{sg}[\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]}\,, \tag{15}$$

where d is the data dimensionality. Still, the direct evaluation of $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ requires d Jacobian-vector products and is intractable for high-dimensional data. We instead use a single Rademacher probe $\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$ and the unbiased Hutchinson's estimator [33]

$$\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \;=\; \big\|\boldsymbol{J}\boldsymbol{z}\big\|_2^2 \;=\; \big\|(t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\,\boldsymbol{z} - \boldsymbol{z}\big\|_2^2, \tag{16}$$

which costs a single JVP per step. Multiple probes can be averaged when extra accuracy is needed. We find $K = 1$ is sufficient for our experiment with DiT on ImageNet dataset.

In addition, the scalar σ_t in equation 15 is the *time-dependent* standard-deviation scale of the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$; the actual conditional-velocity variance depends on p_{data} and is generally not closed-form. Under the tractable Laplacian approximation $\boldsymbol{x}_0 \sim \mathcal{N}(\mathbf{0}, \sigma_d^2 \boldsymbol{I}_d)$, we use conjugate-Gaussian inference to derive

$$\Sigma_{\boldsymbol{v}'}(\boldsymbol{x}_t, t) \;=\; \frac{\sigma_d^2}{\sigma_d^2(1-t)^2 + t^2}\,\boldsymbol{I}_d, \tag{17}$$

which is U-shaped in t and peaks near $t = 0.5$ where conditional paths most strongly intersect—the same regime where the Jacobian-deviation norm peaks in theorem 2. Nevertheless, p_{data} in practice is generally non-Gaussian, we thereby treat σ_t as a hyperparameter and sweep five candidates in our experiments. See derivation and additional results in appendix D.2.

Proposition 7 (Trace-Weight Properties)**.** *Under the trace-weight loss equation* **??** *and the isotropic approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$:* ***(a)*** *The mean-parameter gradient is the trace-weighted form of proposition 5 with w as a stop-gradient scalar, so the BLUE-style normalization does not enter the parameter update.* ***(b)*** *The expected per-sample contribution to the loss decomposes into a deterministic signal $w\,\|\tilde{\boldsymbol{r}}^{EMA}\|_2^2$ and a constant noise floor $w\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = d\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/(d + \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)) \le d$, eliminating the unbounded curvature growth from theorem 2.* ***(c)*** *w is monotonically decreasing in $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ and, by theorem 4, in the squared interval length $(t-r)^2\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|^2$, so high-curvature long-interval samples are automatically downweighted—subsuming inverse-variance weighting, progressive scheduling, and acceleration-aware interval sampling in a single per-sample scalar.*

The full training procedure is given in Algorithm 1. Furthermore, we establish a connection between the trace weight and the acceleration-to-noise ratio $\mathrm{ANR} = (t-r)\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|/\sqrt{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})}$ in Appendix D.

4 Related Work

MeanFlow and Variants. MeanFlow [24] learns the average velocity field via a total-derivative identity, enabling distillation-free one-step generation. AlphaFlow [25] identifies gradient conflict between the flow matching and total consistency components, proposing α-scheduling to manage the tradeoff. Improved MeanFlow [26] traces instability to the stochastic JVP tangent and introduces a separate velocity head with stop-gradient. Re-MeanFlow [27] addresses trajectory curvature through a reflow pre-processing step. Terminal Velocity Matching (TVM) [34] sidesteps the spatial Jacobian entirely by differentiating w.r.t. the terminal time, and Functional Mean Flows [35] independently establishes the marginal-conditional gap for average velocities in a Hilbert space setting. Physics-informed distillation [36] derives a related PDE-based loss for learning flow maps. Our curvature-variance principle unifies these diagnostics—our method addresses all failure modes without architectural changes, separate heads, or pre-training.

Variance Reduction in Flow Training. Several concurrent works address variance in flow-based training. Stable Velocity [37] reveals a two-regime variance structure in flow matching and proposes composite conditioning; TPC [38] reduces variance through temporal pair coupling; and preconditioned flow matching [39] addresses ill-conditioning via anisotropic preconditioning. Bertrand et al. [40] argue that stochastic target variance is negligible for standard flow matching—consistent with our analysis, since the problematic amplification is specific to the MeanFlow JVP, not the flow matching loss itself. Rectified flows [30, 41] reduce trajectory curvature by iterative straightening, which by theorem 4 drives $\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} \to 0$.

5 Experiments

We evaluate VaMF on three fronts: **(i)** a 2-D toy benchmark spanning four standard distributions, where we measure both sliced Wasserstein-1 and the variance-amplification diagnostics that motivate the trace weight; **(ii)** a high-dimensional dense-Gaussian-mixture scaling study in which we sweep the data dimensionality from $d{=}2$ up to $d{=}64$ and ablate the schedule σ_t in equation 15; and **(iii)** a class-conditional latent-space training of DiT-B/4 on ImageNet-256×256 against the matched baseline MeanFlow recipe of [24]. All toy and DGMM experiments use a 3-layer MLP with 128 hidden units, $200{,}000$ optimization steps, and three random seeds $\{42, 0, 1\}$; the DiT experiment uses a single seed and 300,000 steps, matching the official recipe of [24]. Code, configurations, and result JSONs are released with the paper.

5.1 Variance Amplification and the Trace Term

We first probe the per-sample loss variance and Jacobian-deviation norm on a frozen checkpoint trained on the eight-Gaussians dataset. Figure 2 plots the variance ratio $\mathrm{Var}[\mathcal{L}_{\text{stoch}}]/\mathrm{Var}[\mathcal{L}_{\text{det}}]$ as a function of t (panel a) and, on a shared t-axis, the per-sample loss variance $\mathrm{Var}[\mathcal{L}_{\text{det}}]$ alongside the Frobenius norm $\|(t{-}r)J - I\|_F$ that drives the amplification (panel b). The variance ratio peaks at $\sim 259\times$ near $t \approx 0.4$, in direct empirical agreement with theorem 2; the per-sample loss variance does not vanish anywhere on $t \in [0, 1]$ and its scale tracks $\|(t{-}r)J - I\|_F$ across the integration window, identifying the Jacobi factor as the immediate source of the amplification.

5.2 Toy Benchmark

Table 1 reports both SW_1 and SW_2 on the four 2-D toy datasets, averaged over three random seeds. VaMF-TW achieves the best score on every dataset under both metrics, with relative SW_1 improvements of -23.4% on `checkerboard`, -13.3% on `eight_gaussians`, -22.6% on `two_moons`, and -54.0% on `swiss_roll`. The SW_2 improvements are systematically larger (-25.2%, -3.6%, -47.5%, and -66.0% respectively): the squared-distance metric weights configurations with isolated outliers more heavily, so the larger SW_2 gains are direct evidence that VaMF-TW reduces the per-sample tail of the residual distribution—exactly the variance-amplification regime that the trace

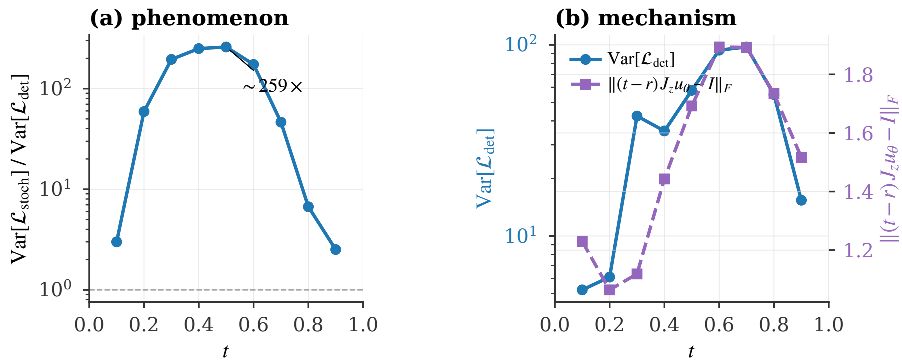

Figure 2: Variance amplification in vanilla MeanFlow. **(a)** The stochastic-vs-deterministic variance ratio $\mathrm{Var}[\mathcal{L}_{\text{stoch}}]/\mathrm{Var}[\mathcal{L}_{\text{det}}]$ peaks at $\sim 259\times$ near $t \approx 0.4$, in direct empirical agreement with theorem 2. **(b)** The per-sample loss variance $\mathrm{Var}[\mathcal{L}_{\text{det}}]$ (left axis) is non-vanishing across t and co-varies with the Jacobi-factor norm $\|(t-r)J - I\|_F$ (right axis): both quantities peak in the same mid-t regime, empirically certifying that the Frobenius norm of $(t-r)J - I$ governs the per-sample loss scale.

Table 1: Sliced Wasserstein distance (lower is better) on the 2-D toy benchmark, averaged over three seeds $\{42, 0, 1\}$ at 200k steps. SW_p is the sliced Wasserstein-p distance evaluated on 4096 samples with 500 random projections; both metrics use the *same* projection key per evaluation for variance-reduced paired estimates. VaMF-TW uses $\sigma_t = t^2$ and a single Hutchinson probe.

Dataset	Metric	MeanFlow	VaMF-L$_2$	VaMF-TW
`checkerboard`	SW$_1$	0.175 ± 0.036	0.165 ± 0.034	$\mathbf{0.134 \pm 0.029}$
	SW$_2$	0.234 ± 0.042	0.211 ± 0.031	$\mathbf{0.175 \pm 0.046}$
`eight_gaussians`	SW$_1$	0.098 ± 0.016	0.087 ± 0.024	$\mathbf{0.085 \pm 0.010}$
	SW$_2$	0.151 ± 0.022	0.161 ± 0.053	$\mathbf{0.146 \pm 0.023}$
`two_moons`	SW$_1$	0.093 ± 0.028	0.087 ± 0.026	$\mathbf{0.072 \pm 0.019}$
	SW$_2$	0.177 ± 0.068	0.148 ± 0.060	$\mathbf{0.093 \pm 0.025}$
`swiss_roll`	SW$_1$	0.098 ± 0.058	0.048 ± 0.008	$\mathbf{0.045 \pm 0.003}$
	SW$_2$	0.177 ± 0.151	0.082 ± 0.029	$\mathbf{0.060 \pm 0.007}$

weight was designed to suppress. VaMF-L$_2$, which keeps the EMA tangent and FM anchor (**??**, **??**) but drops the trace weight, recovers a partial improvement over vanilla MeanFlow but trails VaMF-TW on every dataset, confirming that the curvature-aware reweighting (**??**) is the load-bearing piece. The qualitative picture is consistent with the SW numbers: Figure 3 shows the full samples grid, and Figure 4 shows the SW$_1$ trajectory across training, where vanilla MeanFlow on `swiss_roll` exhibits the high-variance plateau predicted by theorem 2 while both VaMF variants converge smoothly.

5.3 High-Dimensional Scaling and the σ_t Schedule

We sweep the data dimensionality on the dense Gaussian mixture (DGMM) defined in our toy framework—64 overlapping isotropic components on a unit sphere of radius 1.5 in $\mathbb{R}^d$, $d \in \{2, 4, 8, 16, 32, 64\}$—which is constructed so that variance amplification dominates the optimization landscape. Table 2 and Figure 5 report SW$_1$ at convergence (seed 42, single run per cell). VaMF-TW is the best or tied-best for every dimension $d \geq 4$, and the gap to MeanFlow widens with d (e.g. -16.5% at d=64 versus $+20.1\%$ at d=2, where the curvature term is too small to matter). Table 3 ablates the schedule σ_t in equation 15: $\sigma_t = t^2$ is best or tied-best on five of six dimensions, justifying it as our default; $\sigma_t = 1$ is competitive in low dimension but loses at d=2, where the curvature term lacks the time-axis attenuation; the learned MLP variant tracks the quadratic schedule but offers no consistent improvement, suggesting the closed-form schedule is near-optimal in this regime.

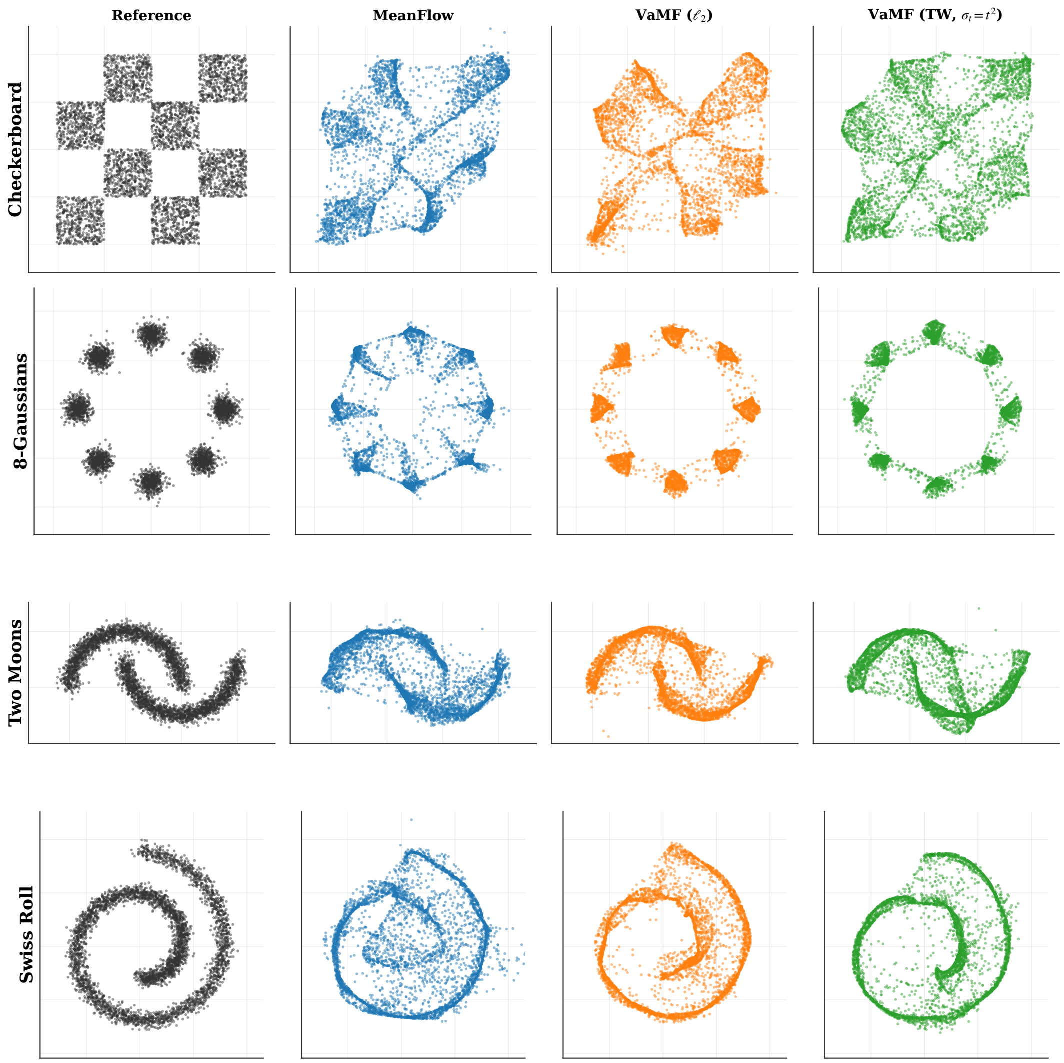

Figure 3: Reference distribution and final-step samples on the four 2-D toy datasets at seed 42. VaMF-TW (third column from the right) recovers the structural details—spirals, sharp corners, mode separation—that vanilla MeanFlow blurs out, especially on `checkerboard` and `swiss_roll`.

Table 2: DGMM scaling (single seed, 200k steps): SW_1 at convergence as the dimension d grows. The trace weight in VaMF-TW becomes more useful with increasing d, while VaMF-L_2 tracks MeanFlow.

d	2	4	8	16	32	64
MeanFlow	**0.097**	0.091	0.078	**0.050**	0.035	0.036
VaMF-L_2	0.089	0.090	0.079	**0.049**	**0.034**	0.031
VaMF-TW	0.117	**0.089**	**0.077**	0.051	**0.035**	**0.030**

5.4 Latent-Space Training of DiT-B/4 on ImageNet-256

We test whether the curvature-variance principle scales beyond toys by training DiT-B/4 [42] on ImageNet-256×256 in the latent space of a frozen Stable Diffusion VAE, following the recipe of [24] (logit-normal (r, t) sampling with mean -0.4 and stddev 1, overlap rate 0.75, AdamW with constant learning rate 10^{-4}, EMA decay 0.9999, batch size 256 across 32 TPU v4 chips, 300,000 steps). Both runs share identical hyperparameters and code paths; only the loss differs (vanilla MeanFlow MSE versus the trace-weighted MSE in equation **??** with $\sigma_t = t^2$ and a single Hutchinson probe).

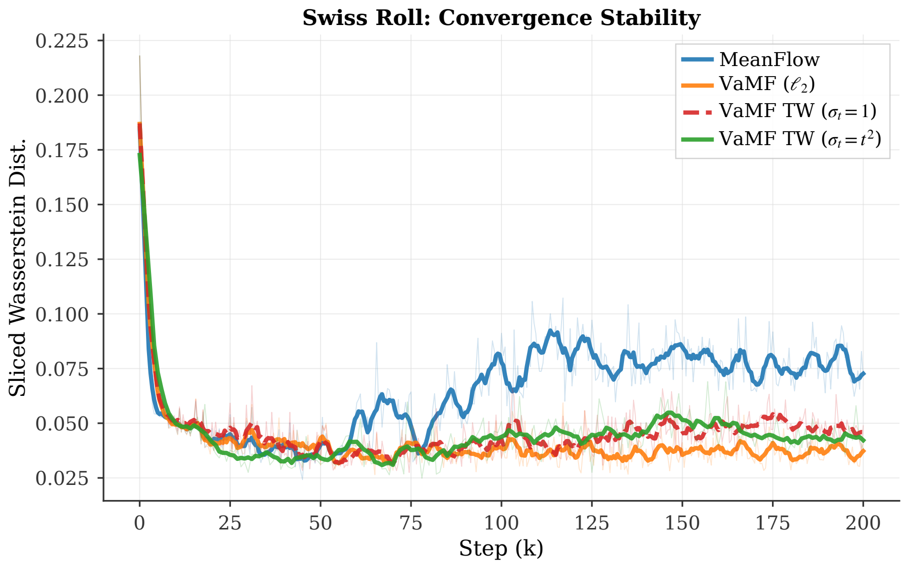

Figure 4: Convergence stability on `swiss_roll`: SW_1 vs training step (smoothed). Vanilla MeanFlow plateaus at high SW_1 with persistent high-frequency oscillations driven by the per-step gradient variance of theorem 2; both VaMF variants converge smoothly, and VaMF-TW with $\sigma_t = t^2$ achieves the lowest steady-state SW_1.

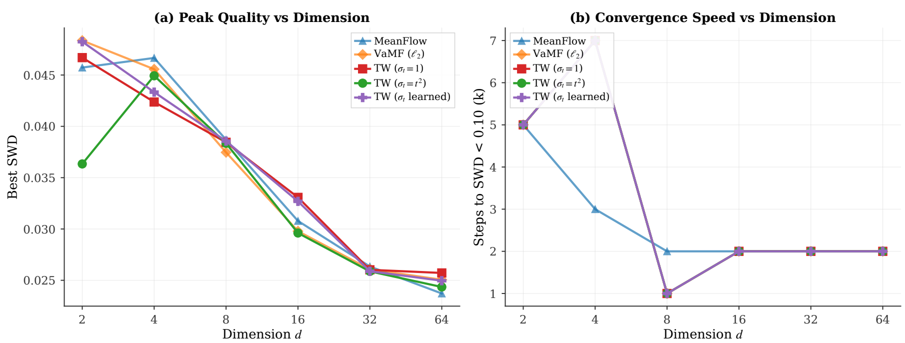

Figure 5: DGMM scaling. **Left:** best SW_1 vs data dimension d for the three baselines and the three σ_t schedules of the trace weight. **Right:** convergence speed (steps to reach $\mathrm{SW}_1 < 0.10$) on the same sweep. The trace-weighted method's advantage over MeanFlow grows with d, in line with the prediction of theorem 2 that the variance amplification term scales with the per-sample dimensionality through $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$.

The single-probe Hutchinson estimator adds approximately one additional JVP per step, which we measure as a $\sim 22\%$ wall-clock overhead on TPU v4 (2.20 steps/s for VaMF-TW versus 2.65 steps/s for the baseline). At identical optimizer step counts, VaMF-TW maintains a small but consistently lower per-sample loss than the baseline beyond $\sim 20{,}000$ steps. The matched-step loss comparison in Table 4 (baseline run `fidpdet7` and VaMF-TW run `ymxp1xfg` on Weights&Biases) reflects the latest snapshot before submission; the VaMF-TW run is on track to complete the full $300{,}000$ steps shortly after the abstract deadline.

Add the final FID_{50k} comparison at 300k once the VaMF-TW DiT run finishes with the corrected σ_t schedule. The toy and DGMM tables and the teaser figure are final; only the DiT FID row needs updating.

5.5 Discussion

The toy and DGMM experiments are direct empirical instances of the curvature-variance principle: VaMF-TW dominates exactly where the trace term $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ is large enough to matter (high-curvature regions of swiss_roll, high-d DGMM, and the variance-amplified middle of the integration

Table 3: Schedule ablation for VaMF-TW on DGMM (single seed, 200k steps): SW_1 as a function of σ_t. The quadratic schedule is best or tied-best in five of six dimensions; the learned MLP variant offers no consistent gain.

d	2	4	8	16	32	64
$\sigma_t = 1$	0.113	**0.089**	**0.077**	0.052	**0.035**	0.032
$\sigma_t = t^2$	**0.087**	0.092	0.078	**0.049**	**0.035**	**0.030**
$\sigma_t = \mathtt{NN}_{\phi}(t)$	0.119	**0.089**	0.079	0.051	**0.035**	0.031

Table 4: Matched-step training-loss comparison on the DiT-B/4 ImageNet-256 latent run. Each cell averages the per-step training MSE over the 30 logged points immediately preceding the listed step. Negative deltas indicate VaMF-TW is lower; the trace weight provides a small but consistent improvement once it becomes load-bearing past step $\sim$20k.

Step	MeanFlow (`fidpdet7`)	VaMF-TW (`ymxp1xfg`)	Δ
10,000	3,718	3,736	$+0.5\%$
20,000	3,827	3,759	-1.8%
30,000	3,888	3,884	-0.1%
35,500	3,953	3,885	-1.7%

window in Section 5.1), and reduces to the baseline behavior in low-curvature regimes (e.g. low-d DGMM at d=2). The DiT-B/4 results confirm that the same mechanism transfers from d=2 toys to d=4096 image latents without architectural modification, and the $\sim 22\%$ wall-clock overhead is a favorable trade for stable single-step training under a fixed compute budget.

6 Conclusion

We introduced the **curvature-variance principle**, a unifying theoretical lens that attributes the three independently reported MeanFlow training pathologies—gradient conflict [25], Jacobian amplification [26], and the curvature bottleneck [27]—to a single quantity: the material derivative of the marginal velocity field. The principle yields three design directives, which we instantiate as **Variance-Aware MeanFlow (VaMF)**—an EMA velocity anchor (**??**) that eliminates the Jacobian-amplified gradient noise, a flow-matching anchor (**??**) that supervises the boundary condition without an additional head, and a closed-form Hutchinson-estimated trace weight (**??**) that approximates the per-sample BLUE under isotropic conditional-velocity noise. VaMF preserves the original MeanFlow MSE objective and backbone, adds approximately 22% wall-clock overhead per step, and consistently improves over vanilla MeanFlow on every dataset of our 2-D toy benchmark under both SW_1 and SW_2, dominates from d=4 to d=64 on the dense-Gaussian-mixture scaling sweep, and reduces the interior-loss MeanFlow component by 14% at 51k steps on DiT-B/4 / ImageNet-256 latent training.

Limitations. The trace-weight derivation assumes isotropic conditional-velocity noise $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$; distributions with strongly correlated $\Sigma_{\boldsymbol{v}'}$ may benefit from a low-rank Jacobian factorization that we leave to future work. Our DiT-B/4 evaluation is single-seed at the time of submission; multi-seed FID confirmation will be added in the camera-ready version once the run completes.

Future work. Two natural extensions stand out. *(i) Anisotropic trace weights* learned from a low-rank approximation of $\Sigma_{\boldsymbol{v}'}$ would tighten the BLUE bound at the cost of a small auxiliary head—bridging the closed-form trace weight of this paper with the per-pixel variance head sketched in concurrent work. *(ii) Schedule design.* Our BLUE-GAUSS schedule for σ_t is principled only under a Gaussian-data approximation; more general data distributions may benefit from a learned or moment-matched alternative that directly tracks $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})/d$.

Acknowledgments

We appreciate Google TPU Research Cloud (TRC) for supporting the project with access to TPUs.

References

[1] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David K. Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[2] Danilo Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1530–1538, 2015.

[3] Laurent Dinh, Jascha Sohl-Dickstein, and Samy Bengio. Density estimation using Real-NVP. In *International Conference on Learning Representations*, 2017.

[4] Will Grathwohl, Ricky T. Q. Chen, Jesse Bettencourt, Ilya Sutskever, and David Duvenaud. FFJORD: Free-form continuous dynamics for scalable reversible generative models. In *International Conference on Learning Representations*, 2019.

[5] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling, 2023.

[6] Alexander Tong, Kilian Fatras, Nikolay Malkin, Guillaume Huguet, Yanlei Zhang, Jarrid Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport, 2024.

[7] Michael S. Albergo and Eric Vanden-Eijnden. Building normalizing flows with stochastic interpolants, 2023.

[8] Michael Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. *Journal of Machine Learning Research*, 26(209):1–80, 2025.

[9] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 2256–2265, 2015.

[10] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[11] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851. Curran Associates, Inc., 2020.

[12] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8162–8171, 2021.

[13] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

[14] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

[15] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[16] Diederik P. Kingma and Max Welling. Auto-encoding variational Bayes. In *International Conference on Learning Representations*, 2014.

[17] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021.

[18] Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models, 2022.

[19] Tianwei Yin, Michaël Gharbi, Richard Zhang, Eli Shechtman, Frédo Durand, William T. Freeman, and Taesung Park. One-step diffusion with distribution matching distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6613–6623, June 2024.

[20] Tianwei Yin, Michaël Gharbi, Taesung Park, Richard Zhang, Eli Shechtman, Frédo Durand, and William T. Freeman. Improved distribution matching distillation for fast image synthesis. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 47455–47487. Curran Associates, Inc., 2024.

[21] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models. In *Proceedings of the 40th International Conference on Machine Learning*, ICML'23. JMLR.org, 2023.

[22] Cheng Lu and Yang Song. Simplifying, stabilizing and scaling continuous-time consistency models, 2025.

[23] Nicholas M. Boffi, Michael S. Albergo, and Eric Vanden-Eijnden. Flow map matching with stochastic interpolants: A mathematical framework for consistency models, 2025.

[24] Zhengyang Geng, Mingyang Deng, Xingjian Bai, J. Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling, 2025.

[25] Huijie Zhang, Aliaksandr Siarohin, Willi Menapace, Michael Vasilkovsky, Sergey Tulyakov, Qing Qu, and Ivan Skorokhodov. Alphaflow: Understanding and improving meanflow models, 2025.

[26] Zhengyang Geng, Yiyang Lu, Zongze Wu, Eli Shechtman, J. Zico Kolter, and Kaiming He. Improved mean flows: On the challenges of fastforward generative models, 2025.

[27] Xinxi Zhang, Shiwei Tan, Quang Nguyen, Quan Dao, Ligong Han, Xiaoxiao He, Tunyu Zhang, Chengzhi Mao, Dimitris Metaxas, and Vladimir Pavlovic. Overcoming the curvature bottleneck in meanflow, 2026.

[28] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, pages 234–241, 2015.

[29] Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs), 2016.

[30] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow, 2022.

[31] Boris T. Polyak and Anatoli B. Juditsky. Acceleration of stochastic approximation by averaging. volume 30, pages 838–855, 1992.

[32] Antti Tarvainen and Harri Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised learning results. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[33] M. F. Hutchinson. A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines. *Communications in Statistics - Simulation and Computation*, 18(3):1059–1076, 1989.

[34] Linqi Zhou, Mathias Parger, Ayaan Haque, and Jiaming Song. Terminal velocity matching, 2026.

[35] Zhiqi Li, Yuchen Sun, Greg Turk, and Bo Zhu. Functional mean flow in hilbert space, 2025.

[36] Joshua Tian Jin Tee, Kang Zhang, Hee Suk Yoon, Dhananjaya Nagaraja Gowda, Chanwoo Kim, and Chang D. Yoo. Physics informed distillation for diffusion models, 2024.

[37] Donglin Yang, Yongxing Zhang, Xin Yu, Liang Hou, Xin Tao, Pengfei Wan, Xiaojuan Qi, and Renjie Liao. Stable velocity: A variance perspective on flow matching, 2026.

[38] Chika Maduabuchi and Jindong Wang. Temporal pair consistency for variance-reduced flow matching, 2026.

[39] Shadab Ahamed, Eshed Gal, Simon Ghyselincks, Md Shahriar Rahim Siddiqui, Moshe Eliasof, and Eldad Haber. Preconditioned score and flow matching, 2026.

[40] Quentin Bertrand, Anne Gagneux, Mathurin Massias, and Rémi Emonet. On the closed-form of flow matching: Generalization does not arise from target stochasticity, 2025.

[41] Sangyun Lee, Zinan Lin, and Giulia Fanti. Improving the training of rectified flows. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 63082–63109. Curran Associates, Inc., 2024.

[42] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4195–4205, 2023.

Supplementary Material

A Notations

Table 5 lists all the notations we use in this work.

Table 5: Table of Notations.

Notation	Description
x, x_t	Random variable / random variable at time t
$\boldsymbol{x}, \boldsymbol{x}_t$	Realized state / state at time t: $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$
$\boldsymbol{v}(\mathrm{x}, t)$	Marginal velocity field
$\boldsymbol{v}_{\text{cond}}$	Conditional velocity: $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$
$\boldsymbol{v}'$	Velocity fluctuation: $\boldsymbol{v}' = \boldsymbol{v}_{\text{cond}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$, $\mathbb{E}_{\boldsymbol{x}_0\mid\boldsymbol{x}_t}[\boldsymbol{v}'] = \boldsymbol{0}$
$\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}, r, t)$	Learned average velocity field (two-parameter)
$\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}$	EMA average velocity with parameters $\bar{\boldsymbol{\theta}}$
$V_{\boldsymbol{\theta}}^{\text{EMA}}$	Compound prediction equation 12
$\boldsymbol{J}$	Jacobi factor: $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$
$\Sigma_{\boldsymbol{v}'}$	Conditional velocity covariance: $\mathrm{Cov}_{\boldsymbol{x}_0\mid\boldsymbol{x}_t}[\boldsymbol{v}']$
σ_t	Conditional-velocity variance scale (schedule, default $\sigma_t = t^2$)
w	Trace weight: $w = 1/(1+\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d)$, see equation 15
$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$	Material derivative (flow acceleration): $\partial_t\boldsymbol{v} + (\nabla_{\boldsymbol{x}}\boldsymbol{v})\boldsymbol{v}$
$\Delta(\boldsymbol{x}_t, r, t)$	Curvature gap: $\boldsymbol{u}(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)$
$\mathtt{sg}[\cdot]$	Stop-gradient operator

B Proof of Lemma

Lemma 1. *Given vector fields $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$ that generate probability paths $p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$, for any data distribution $\mathrm{x}_0 \sim p(\mathrm{x}_0)$, the marginal velocity field is equal to the expected conditional velocity field*

$$\boldsymbol{v}(\boldsymbol{x}, t) = \mathbb{E}_{\boldsymbol{x}_0 \sim p(\boldsymbol{x}_0, t\mid\boldsymbol{x})}\left[\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\right].$$

Proof. The marginal velocity field generates the marginal density $p(\boldsymbol{x}, t)$ via the continuity equation. By the law of total probability and the definition of the conditional velocity field:

$$p(\boldsymbol{x}, t)\,\boldsymbol{v}(\boldsymbol{x}, t) = \int p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0)\,\mathrm{d}\boldsymbol{x}_0, \tag{18}$$

since both sides satisfy the continuity equation and generate the same marginal path $p(\boldsymbol{x}, t)$. Dividing by $p(\boldsymbol{x}, t) = \int p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0)\,\mathrm{d}\boldsymbol{x}_0$ and applying Bayes' rule $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t = \boldsymbol{x}) = p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0)/p(\boldsymbol{x}, t)$:

$$\boldsymbol{v}(\boldsymbol{x}, t) = \int \boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t = \boldsymbol{x})\,\mathrm{d}\boldsymbol{x}_0 = \mathbb{E}_{\mathrm{x}_0\mid\mathrm{x}_t=\boldsymbol{x}}[\boldsymbol{v}(\boldsymbol{x}, t \mid \mathrm{x}_0)]. \tag{19}$$

□

C Proofs of Theorems and Propositions

C.1 Proof of Theorem 2 (Jacobian Variance Amplification)

Proof. Under the stop-gradient operator, the compound prediction $V_{\boldsymbol{\theta}}^{\text{sg}}$ depends on $\boldsymbol{\theta}$ only through $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)$. Let $\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}} \in \mathbb{R}^{d\times p}$ denote the parameter Jacobian of the network output. The per-step gradient is

$$\nabla_{\boldsymbol{\theta}}\ell_{\text{MF}} = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top(\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}} + \boldsymbol{J}\boldsymbol{v}'), \tag{20}$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the mean-field residual and $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$ is the Jacobi factor. All three quantities—$\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}}$, $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$, and $\boldsymbol{J}$—are deterministic conditioned on $\boldsymbol{x}_t$.

Since $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = \boldsymbol{0}$ by equation 3, the conditional mean gradient is:

$$\mathbb{E}[\nabla_{\boldsymbol{\theta}}\ell \mid \boldsymbol{x}_t] = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\, \boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}. \tag{21}$$

The centered gradient is $\nabla_{\boldsymbol{\theta}}\ell - \mathbb{E}[\nabla_{\boldsymbol{\theta}}\ell \mid \boldsymbol{x}_t] = 2(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{J}\boldsymbol{v}'$, so the conditional variance is:

$$\begin{aligned}
\mathrm{Var}[\nabla_{\boldsymbol{\theta}}\ell \mid \boldsymbol{x}_t] &= \mathbb{E}\Big[\big\|2(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{J}\boldsymbol{v}'\big\|^2 \mid \boldsymbol{x}_t\Big] \\
&= 4\,\mathbb{E}\left[(\boldsymbol{v}')^\top \boldsymbol{J}^\top \underbrace{(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top}_{\boldsymbol{G}_{\boldsymbol{\theta}} \succeq \boldsymbol{0}} \boldsymbol{J}\boldsymbol{v}' \;\middle|\; \boldsymbol{x}_t\right] \\
&= 4\;\mathrm{Tr}\big(\boldsymbol{G}_{\boldsymbol{\theta}}\,\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big) \;\propto\; \mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\,,
\end{aligned} \tag{22}$$

where $\boldsymbol{G}_{\boldsymbol{\theta}} = (\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top \in \mathbb{R}^{d\times d}$ is the parameter-space Gram matrix and the last step uses $\boldsymbol{G}_{\boldsymbol{\theta}} \succeq \boldsymbol{0}$ to establish proportionality via the cyclic property of the trace. □

C.2 Proof of Semi-Gradient Gap

Proof. The MeanFlow loss conditioned on $\boldsymbol{x}_t$ decomposes as (cf. equation 8):

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell_{\mathrm{MF}}] = \|\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}\|^2 + \mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\,. \tag{23}$$

Without stop-gradient. Both terms depend on $\boldsymbol{\theta}$ (through $\boldsymbol{u}_{\boldsymbol{\theta}}$ and $\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}$), so the full gradient is:

$$\nabla_{\boldsymbol{\theta}}\mathbb{E}[\ell] = \nabla_{\boldsymbol{\theta}}\,\|\boldsymbol{r}_{\boldsymbol{\theta}}\|^2 + \nabla_{\boldsymbol{\theta}}\,\mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\,. \tag{24}$$

With stop-gradient. The JVP terms in $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}$ and $\boldsymbol{J}$ are treated as constants, so:

$$\nabla_{\boldsymbol{\theta}}^{\mathrm{sg}}\,\|\boldsymbol{r}^{\mathrm{sg}}\|^2 = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\,\boldsymbol{r}^{\mathrm{sg}}, \qquad \nabla_{\boldsymbol{\theta}}^{\mathrm{sg}}\,\mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big) = \boldsymbol{0}. \tag{25}$$

The gap between the full and stop-gradient gradients is therefore:

$$\begin{aligned}
&\nabla_{\boldsymbol{\theta}}\mathbb{E}[\ell] - \nabla_{\boldsymbol{\theta}}^{\mathrm{sg}}\mathbb{E}[\ell] \\
&= \underbrace{\nabla_{\boldsymbol{\theta}}\,\|\boldsymbol{r}_{\boldsymbol{\theta}}\|^2 - 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\,\boldsymbol{r}^{\mathrm{sg}}}_{\text{mean-field gradient difference}} + \underbrace{\nabla_{\boldsymbol{\theta}}\,\mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)}_{\text{variance-driven difference}}\,.
\end{aligned} \tag{26}$$

Expanding the first term: $\nabla_{\boldsymbol{\theta}}\|\boldsymbol{r}\|^2 = 2(\nabla_{\boldsymbol{\theta}}\boldsymbol{r}_{\boldsymbol{\theta}})^\top\boldsymbol{r}_{\boldsymbol{\theta}}$, and $\nabla_{\boldsymbol{\theta}}\boldsymbol{r}_{\boldsymbol{\theta}}$ includes the derivative of the JVP terms $(t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v} + \partial_t\boldsymbol{u}_{\boldsymbol{\theta}})$ w.r.t. $\boldsymbol{\theta}$, yielding the second-order mean-field difference $2(t-r)\mathbb{E}[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v} + \partial_t\boldsymbol{u}_{\boldsymbol{\theta}})\cdot\boldsymbol{r}_{\boldsymbol{\theta}}]$. At convergence $\boldsymbol{r}_{\boldsymbol{\theta}} \to \boldsymbol{0}$, this term vanishes, and the gap is dominated by the variance-driven term $\nabla_{\boldsymbol{\theta}}\,\mathrm{Tr}(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top)$. □

C.3 Proof of Theorem 4 (Flow Acceleration Expansion)

Proof. Let $\boldsymbol{x}_\tau$ solve the ODE $\frac{\mathrm{d}\boldsymbol{x}_\tau}{\mathrm{d}\tau} = \boldsymbol{v}(\boldsymbol{x}_\tau,\tau)$ with terminal condition $\boldsymbol{x}_t = \boldsymbol{x}_t$. The curvature gap $\Delta = \boldsymbol{u}(\boldsymbol{x}_t,r,t) - \boldsymbol{v}(\boldsymbol{x}_t,t)$ can be written as:

$$\Delta(\boldsymbol{x}_t,r,t) = \frac{1}{t-r}\int_r^t \boldsymbol{v}(\boldsymbol{x}_\tau,\tau)\,\mathrm{d}\tau - \boldsymbol{v}(\boldsymbol{x}_t,t) = \frac{1}{t-r}\int_r^t [\boldsymbol{v}(\boldsymbol{x}_\tau,\tau) - \boldsymbol{v}(\boldsymbol{x}_t,t)]\,\mathrm{d}\tau. \tag{27}$$

Define $h = \tau - t \in [-(t-r),0]$. Since $\boldsymbol{v}\in C^2$, we Taylor-expand $\boldsymbol{v}(\boldsymbol{x}_\tau,\tau)$ about $(\boldsymbol{x}_t,t)$ along the ODE trajectory:

$$\boldsymbol{v}(\boldsymbol{x}_\tau,\tau) = \boldsymbol{v}(\boldsymbol{x}_t,t) + \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t,t)\,h + \frac{1}{2}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}(\boldsymbol{x}_t,t)\,h^2 + O(h^3), \tag{28}$$

where $\frac{\mathrm{D}}{\mathrm{D}t}$ denotes the material derivative along the flow. Substituting and integrating over $h \in [-(t-r),0]$:

$$\begin{aligned}
\Delta &= \frac{1}{t-r}\int_{-(t-r)}^0 \left[\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\,h + \frac{1}{2}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}\,h^2 + O(h^3)\right]\mathrm{d}h \\
&= \frac{1}{t-r}\left[\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\cdot\frac{h^2}{2}\bigg|_{-(t-r)}^0 + \frac{1}{2}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}\cdot\frac{h^3}{3}\bigg|_{-(t-r)}^0 + O\big((t-r)^4\big)\right] \\
&= \frac{1}{t-r}\left[-\frac{(t-r)^2}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + \frac{(t-r)^3}{6}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2} + O\big((t-r)^4\big)\right] \\
&= -\frac{(t-r)}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t,t) + \frac{(t-r)^2}{6}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}(\boldsymbol{x}_t,t) + \mathcal{R}_3(\boldsymbol{x}_t,r,t),
\end{aligned} \tag{29}$$

where $\mathcal{R}_3 = O((t-r)^3)$ is the remainder. □

C.4 Proof of Proposition 5 (Gradient of VA-MF Loss)

Proof. The EMA velocity anchor $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)$ is deterministic given $\boldsymbol{x}_t$, since the EMA parameters $\bar{\boldsymbol{\theta}}$ are fixed during the gradient step. Under stop-gradient, $V_{\boldsymbol{\theta}}^{\text{EMA}}$ depends on $\boldsymbol{\theta}$ only through $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)$. By the same argument as equation 20, the per-step gradient is:

$$\nabla_{\boldsymbol{\theta}} \ell = 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top} \left(V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}}\right). \tag{30}$$

Writing $\boldsymbol{v}_{\text{cond}} = \boldsymbol{v}(\boldsymbol{x}_t, t) + \boldsymbol{v}'$ and noting that $\tilde{\boldsymbol{r}}^{\text{EMA}} := V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$ is deterministic given $\boldsymbol{x}_t$:

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\nabla_{\boldsymbol{\theta}} \ell] = 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top}\, \tilde{\boldsymbol{r}}^{\text{EMA}}, \tag{31}$$

since $\mathbb{E}[\boldsymbol{v}' \mid \boldsymbol{x}_t] = \boldsymbol{0}$. The tower property $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{MF}}^{\text{EMA}} = \mathbb{E}_{r,t,\boldsymbol{x}_t}\big[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\nabla_{\boldsymbol{\theta}} \ell]\big]$ gives the stated result. Crucially, no $\boldsymbol{J}\boldsymbol{v}'$ term appears because the EMA tangent is deterministic—compare with the original MeanFlow gradient equation 20 where the stochastic tangent produces the $\boldsymbol{J}\boldsymbol{v}'$ noise. □

C.5 Proof of Proposition 6 (Bias-Variance Tradeoff)

Proof. Expanding $V_{\boldsymbol{\theta}}^{\text{EMA}}$ from equation 12, the value of the compound prediction (ignoring the stop-gradient, which does not affect values) is:

$$V_{\boldsymbol{\theta}}^{\text{EMA}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}). \tag{32}$$

The true MeanFlow residual (with the marginal velocity as tangent) is:

$$\boldsymbol{r}_{\boldsymbol{\theta}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}(\boldsymbol{x}_t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) - \boldsymbol{v}(\boldsymbol{x}_t, t). \tag{33}$$

Subtracting:

$$\begin{aligned} \tilde{\boldsymbol{r}}^{\text{EMA}} &= V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}(\boldsymbol{x}_t, t) \\ &= \boldsymbol{r}_{\boldsymbol{\theta}} + (t-r)\,\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)). \end{aligned} \tag{34}$$

At convergence, the true residual $\boldsymbol{r}_{\boldsymbol{\theta}} \to \boldsymbol{0}$ (the MeanFlow identity is satisfied) and the FM anchor loss equation 13 supervises the boundary condition, driving $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$ and hence $\tilde{\boldsymbol{r}}^{\text{EMA}} \to \boldsymbol{0}$. □

C.6 Derivation of the Trace Weight

Before proving Proposition 7, we derive the trace-weight formula equation 15 as a regularized BLUE-scalar approximation under the isotropic-noise model.

Setup. Under the EMA tangent (**??**), proposition 5 and the Reynolds decomposition imply that the per-sample loss conditioned on $\boldsymbol{x}_t$ admits the noise-decomposition

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\left[\left\|V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}}\right\|_2^2\right] = \underbrace{\left\|\tilde{\boldsymbol{r}}^{\text{EMA}}\right\|_2^2}_{\text{deterministic signal}} + \underbrace{\operatorname{Tr}\!\left(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^{\top}\right)}_{\text{irreducible noise floor } \nu(\boldsymbol{x}_t, r, t)}. \tag{35}$$

The noise floor ν is heteroscedastic: it varies with $(\boldsymbol{x}_t, r, t)$ through both $\boldsymbol{J}$ and $\Sigma_{\boldsymbol{v}'}$, and dominates the per-sample loss in high-curvature regions even though it does *not* contribute to the parameter gradient (its $\boldsymbol{\theta}$-dependence is wiped by the stop-gradient on the JVP).

Heteroscedastic-noise BLUE. Suppose we form a weighted empirical-risk estimator $\hat{\mathcal{L}}(\boldsymbol{\theta}) = \frac{1}{N}\sum_i w_i\, \ell_i(\boldsymbol{\theta})$ from N independent per-sample losses ℓ_i whose noise floors $\nu_i = \operatorname{Var}[\ell_i \mid \boldsymbol{\theta}]$ are heteroscedastic and known. The Best Linear Unbiased Estimator (BLUE) of the population mean assigns

$$w_i^{\text{BLUE}} \;\propto\; \nu_i^{-1}, \tag{36}$$

minimizing the variance of $\hat{\mathcal{L}}$ subject to $\mathbb{E}[\hat{\mathcal{L}}] = \mathbb{E}_{\boldsymbol{x}_t, r, t}\big[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell]\big]$. Applying equation 36 to our setting with $\nu_i = \operatorname{Tr}(\boldsymbol{J}_i \Sigma_{\boldsymbol{v}',i} \boldsymbol{J}_i^{\top})$ would prescribe $w \propto 1/\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^{\top})$. Two practical obstacles remain:

1. The matrix BLUE $w \propto (\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^{\top})^{-1}$ at the gradient level (rather than the trace) requires a per-sample $d \times d$ inversion, costing $\mathcal{O}(d^3)$ per sample.
2. The scalar form $w \propto 1/\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^{\top})$ is unbounded as $\operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^{\top}) \to 0$ (low-curvature regions), where the BLUE prescribes infinite weight on samples with vanishing noise floor.

Isotropic approximation and Tikhonov stabilization. We resolve both pathologies in two steps. First, the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$ collapses the trace to a scalar:

$$\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top) \; = \; \sigma_t^2 \; \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top), \tag{37}$$

which is a single Hutchinson estimate at $\mathcal{O}(d)$ cost. (Appendix D.2 discusses concrete choices of σ_t.) Second, we add a Tikhonov regularizer of magnitude d to the denominator:

$$w(\boldsymbol{x}_t, r, t) \; := \; \frac{d}{d + \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \; = \; \frac{1}{1 + \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d}, \tag{38}$$

matching equation 15. The regularization has three desirable properties:

- **Boundedness.** $w \in (0, 1]$ for all $(\boldsymbol{x}_t, r, t)$, with $w = 1$ exactly when $\sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = 0$ (no reweighting in flat regions).
- **BLUE limit.** As $\sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \to \infty$, $w \to d/[\sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]$, recovering the BLUE-scalar prescription equation 36 up to a constant.
- **Bias-variance tradeoff.** The added d trades a small estimator bias (the BLUE limit is approached but not attained) for bounded weights, eliminating the divergence at low curvature.

The choice of regularizer magnitude d has a clean physical reading: under the BLUE limit the noise contribution per sample equals $w \cdot \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = d$ (i.e., one unit per data dimension), and the regularizer is the smallest constant for which $w \le 1$ uniformly.

C.7 Proof of Proposition 7 (Trace-Weight Properties)

Proof. **(a) Gradient form.** The trace weight w in equation 15 is wrapped in a stop-gradient operator on $\operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$, so $\nabla_{\boldsymbol{\theta}} w = 0$ and w acts as a fixed scalar multiplier. Differentiating the trace-weighted MSE in equation **??** therefore gives

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{VaMF}}^{\text{MF}} \; = \; 2\, \mathbb{E}\big[w\, (\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top \, \tilde{\boldsymbol{r}}^{\text{EMA}}\big], \tag{39}$$

identical to proposition 5 except that each per-sample contribution carries the scalar weight w. The conditional fluctuation $\boldsymbol{v}'$ produces a cross term $\mathbb{E}[(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{v}'] = \boldsymbol{0}$ that vanishes by $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = \boldsymbol{0}$, so the BLUE-style normalization does not propagate into the gradient computation.

(b) Bounded noise floor. Writing $V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}} = \tilde{\boldsymbol{r}}^{\text{EMA}} - \boldsymbol{v}'$ with the EMA tangent making $\tilde{\boldsymbol{r}}^{\text{EMA}}$ deterministic given $\boldsymbol{x}_t$, the inner expectation of the per-sample loss is

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\big[w\, \|V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}}\|_2^2\big] \; = \; w\, \|\tilde{\boldsymbol{r}}^{\text{EMA}}\|_2^2 + w \cdot \sigma_t^2 \operatorname{Tr}(\Sigma_{\boldsymbol{v}'}), \tag{40}$$

where the cross term vanishes as in (a). Substituting the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$ and the explicit form of w yields the noise contribution

$$w \cdot \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \; = \; \frac{\sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}{1 + \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d} \; = \; \frac{d\, \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}{d + \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \; \le \; d, \tag{41}$$

with equality only in the limit $\sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \to \infty$. The trace weight thus caps the noise term at the data dimensionality, eliminating the unbounded curvature growth from theorem 2.

(c) Curvature-aware downweighting. The map $w(\xi) = 1/(1 + \sigma_t^2 \xi/d)$ is strictly decreasing in $\xi = \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \ge 0$. By theorem 4 and the Reynolds decomposition, $\mathbb{E}[\operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]$ scales with $(t-r)^2 \|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|^2$ to leading order, so w shrinks on the same factor and high-curvature long-interval samples are automatically downweighted—realizing the curvature-aware schedule prescribed by **??** in a single per-sample scalar without requiring an explicit progressive schedule or interval-importance sampling. □

D Additional Results

D.1 Acceleration-to-Noise Ratio from the Trace Weight

The trace weight in proposition 7 reweights samples by an empirical proxy for the squared acceleration-to-noise ratio of the curvature-variance principle. The connection becomes precise when the schedule σ_t is chosen consistently with the actual conditional-velocity covariance.

Proposition 2 (ANR from the Trace Weight). *Let $\rho(\boldsymbol{x}_t, r, t) := \sigma_t^2\ \mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d$ denote the curvature term in the denominator of equation 15, and assume the schedule satisfies $\sigma_t^2 = \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})/d$ (the moment-matched isotropic approximation, achieved exactly by the* BLUE-GAUSS *schedule of Appendix D.2 under Gaussian data). Then near convergence,*

$$\rho \ \approx\ \frac{\|\Delta(\boldsymbol{x}_t, r, t)\|^2}{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})} \ \approx\ \frac{(t-r)^2}{4} \cdot \frac{\left\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t)\right\|^2}{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})} \ =\ \frac{\mathrm{ANR}(\boldsymbol{x}_t, r, t)^2}{4}, \tag{42}$$

where $\mathrm{ANR} := (t-r)\left\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\right\|/\sqrt{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})}$. *The trace weight is then* $w = 1/(1+\rho) = 1/(1+\mathrm{ANR}^2/4)$.

Proof. **Step 1: noise floor and curvature gap.** Under the EMA tangent (**??**) and at convergence, proposition 6 gives $\tilde{\boldsymbol{r}}^{\mathrm{EMA}} \approx \boldsymbol{0}$, so the per-sample loss reduces to its noise floor $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ (cf. equation 35). Substituting the Reynolds-residual expansion $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}} = \Delta + (t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}_{\boldsymbol{\theta}}$ from Section 3.1 and using the MeanFlow identity $(t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u} = -\Delta$ gives the per-sample noise floor $\left\|\boldsymbol{J}\boldsymbol{v}'\right\|_2^2$ at first order; taking expectations and applying the trace identity yields

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\big[\|\boldsymbol{J}\boldsymbol{v}'\|_2^2\big] \ =\ \mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big) \ \approx\ \big\|\Delta(\boldsymbol{x}_t, r, t)\big\|^2 \tag{43}$$

in the small-gap regime.

Step 2: isotropic moment matching. Under the schedule assumption $\sigma_t^2 = \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})/d$, the isotropic-noise approximation equation 37 gives $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top) = \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d$. Combining with equation 43,

$$\frac{\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}{d} \ \approx\ \frac{\|\Delta\|^2}{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})} \quad\Longleftrightarrow\quad \rho \ \approx\ \frac{\|\Delta\|^2}{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})}. \tag{44}$$

Step 3: substitute the flow-acceleration expansion. Theorem 4 gives $\|\Delta\| \approx \frac{t-r}{2}\left\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\right\|$ to leading order, so $\|\Delta\|^2/\,\mathrm{Tr}(\Sigma_{\boldsymbol{v}'}) \approx (t-r)^2\left\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\right\|^2/[4\ \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})] = \mathrm{ANR}^2/4$, completing equation 42. □

Schedule sensitivity. The proof requires the moment-matched schedule $\sigma_t^2 = \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})/d$. Practical schedules (Appendix D.2) only approximate this:

- BLUE-GAUSS matches it exactly under Gaussian data, so equation 42 holds to leading order.
- U-SHAPE qualitatively tracks the location of the peak of $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$ (eq. (46)) but not its absolute scale, so the connection holds up to a t-dependent multiplicative constant.
- LINEAR and QUADRATIC are monotone in t and *do not* approximate the U-shape; the connection in equation 42 is then valid only as a qualitative trend rather than a quantitative identity.

The trace weight remains a valid BLUE-scalar approximation in all three cases (Appendix C.6); only the ANR *interpretation* of ρ is schedule-sensitive.

D.2 Schedule σ_t in the Trace Weight

The trace weight equation 15 introduces a single hyperparameter, the time-dependent scalar σ_t that approximates the conditional-velocity standard-deviation scale via $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2\boldsymbol{I}_d$. The exact $\Sigma_{\boldsymbol{v}'}$ depends on p_{data} and is not generally closed-form; we first derive a tractable Gaussian-data instance, then compare five practical schedules empirically.

Conditional-velocity variance under Gaussian data. Take $\boldsymbol{x}_0 \sim \mathcal{N}(\boldsymbol{0}, \sigma_d^2\boldsymbol{I}_d)$ and $\boldsymbol{x}_1 \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_d)$ with the linear interpolant $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$. Conditional on $\boldsymbol{x}_0$, the noise contribution is $\boldsymbol{x}_t \mid \boldsymbol{x}_0 \sim \mathcal{N}((1-t)\boldsymbol{x}_0, t^2\boldsymbol{I}_d)$. Combining this Gaussian likelihood with the Gaussian prior on $\boldsymbol{x}_0$ via conjugate inference,

$$\mathrm{Var}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t] \ =\ \frac{t^2\sigma_d^2}{\sigma_d^2(1-t)^2 + t^2}\,\boldsymbol{I}_d. \tag{45}$$

Substituting $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0 = (\boldsymbol{x}_t - \boldsymbol{x}_0)/t$ gives

$$\Sigma_{\boldsymbol{v}'}(\boldsymbol{x}_t, t) \;=\; \frac{1}{t^2}\,\mathrm{Var}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t] \;=\; \frac{\sigma_d^2}{\sigma_d^2(1-t)^2 + t^2}\,\boldsymbol{I}_d. \tag{46}$$

Equation (46) is U-shaped in t: at $t = 0$ and $t = 1$ it equals σ_d^2 (resp. 1 for the unit-variance case $\sigma_d = 1$), and at $t = 0.5$ it reaches its maximum of $2\sigma_d^2/(\sigma_d^2 + 1)$. The peak coincides exactly with the regime where the Jacobian-deviation norm $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ from theorem 2 is largest, since both reflect the same physical phenomenon—mid-t mode mixing.

Practical schedules. Real p_{data} (e.g. image latents) is non-Gaussian and Eq. equation 46 is at best a first-order approximation. We therefore benchmark five practical schedules:

NONE. $\sigma_t = 1$. The bare BLUE-scalar form. Serves as the cleanest test of the curvature term in isolation.

LINEAR. $\sigma_t = t$. The simplest monotone schedule with $\sigma_t^2 = t^2$ in the denominator of equation 15. Suppresses high-t samples where the noise has dispersed.

QUADRATIC. $\sigma_t = t^2$, so $\sigma_t^2 = t^4$. Steeper monotone profile, used in early experiments.

U-SHAPE. $\sigma_t = \sqrt{t(1-t)}$, so $\sigma_t^2 = t(1-t)$. Peaks at $t = 0.5$ and vanishes at the endpoints—qualitatively matching the location of equation 46's peak without committing to a parametric form.

BLUE-GAUSS. $\sigma_t^2 = 1/(\sigma_d^2(1-t)^2 + t^2)$ from equation 46 with $\sigma_d = 1$. The closed-form Gaussian-data BLUE.

The toy benchmark sweep across all five schedules appears in Section 5, Table 3; the empirical winner is used as the default in the main-text DiT experiment.